

Word-count declaration

Manuscript: *A Reproducible Benchmark of Relational Database Access-Layer Performance in Express.js: A Configuration-Specific Comparison on PostgreSQL and MySQL* (submission build `ist/ist_main.tex`).

Counted with `texcount` on the seven body sections plus the structured abstract. Following the IST author guidelines, each table and figure counts as 200 words, and **the reference list is included in the total**.

Component	Count
Body text (7 sections)	11,448
Structured abstract	299
Tables and figures in the main text (7×200)	1,400
Reference list (62 entries)	1,848
Total (IST rule)	14,995

This is under the journal’s 15,000-word limit (with the abstract counted; excluding the abstract it is 14,696). The structured abstract is 299 words (under the journal’s 300-word structured-abstract limit). Thirty-one tables and two figures are placed in the numbered online supplement (`supplement.pdf`, Supplement Tables S1–S31 and Supplement Figures S1–S2), which is submitted with the manuscript and archived under the same Zenodo DOI as the code and data; supplement floats are not part of the main-text count. A later revision added per-cell bootstrap 95% CIs to the p99 column of every pattern table and a paired p99 significance table (Supplement Table S30), delivering the p99 inference the outcomes table lists as primary; these widen existing tables and add one supplement float, not a main-text float.

In this revision (round 5) the entire benchmark was re-run on the latest stable library versions (Prisma 7, Express 5, etc.); the manuscript was repositioned as a reproducible benchmark and a configuration-specific comparison; the body prose was cut by roughly 1,800 words to remove repeated conclusions and filler (the five recurring findings are now stated once, in the results section, and cross-referenced elsewhere); the secondary access patterns (point read, key-set range scan) were reduced to lean summaries with their full tables in the supplement; the controlled/varied/uncontrolled factor table was moved to the supplement (Table S27); and five peripheral citations (general NoSQL and OLTP benchmark context) were dropped, taking the reference list from 70 to 62 entries. Six single-host experiments added earlier in the round (utilization-controlled open-loop, multi-worker cluster, per-layer pool frontier on both engines, mixed read/write, replicated fan-out, and a longer-run sensitivity check) remain as Supplement Tables S21–S26 with a one-sentence pointer each in the body.

Two later patch revisions left the reference list at 62 entries: the re-review revision (v1.5.1) dropped one peripheral citation, and v1.5.2 added one methodological citation (Papadopoulos et al. 2021, *IEEE Transactions on Software Engineering*, on reproducible performance-evaluation principles) anchoring the paper’s perishability thesis. That citation adds no body prose (a bare `\cite`), so the only change to the count is the single reference entry (reference list 1,842 $\rightarrow$ 1,848 words, total 14,917 $\rightarrow$ 14,923).

Revision round 6 retracted the causal same-SQL attribution (now a bounding contrast), renamed the "idiomatic" estimand, separated the capacity/overload/matched-utilization quantities, tempered the perishability claim, and added the required statistical disclosures (Kendall’s τ , an ex-MikroORM robustness note, a layer $\times$ engine interaction table, and an insert-dispersion figure, all in the supplement). The condensed Prisma case study and the removal of repeated restatements offset most of these additions, so the body changed marginally (11,376 to 11,388) and the abstract was tightened from 313 to 298 words; the total is 14,934. The longer-window tail

sensitivity (Table S23), interaction magnitude (Table S28), and insert dispersion (Figure S1) are the new supplement floats.

A subsequent round-6 clarification renamed the primary 50-connection matrix from "saturated" to a "fixed 50-connection high-load operating point," since the throughput knee is demonstrated (Supplement Figure S2) only for the deep fetch, while the ten-connection pool—not a per-pattern knee—is the binding high-load constraint (Section 3, Controls); the per-layer/pool/open-loop "saturat*" uses are unchanged. The rename plus its one-clause justification added a net 15 body words (offset by tightening three tail-latency sentences) and one abstract word, taking the body to 11,444 and the total to 14,991.

A further round-6 clarification softened the causal claim that each layer is "single-core-bound" / "multi-core-bound": the equal-CPU and multi-worker controls show that extra application cores do not raise a process's throughput (so the ranking is not a core-budget artifact), but do not prove single-core CPU is the causal bottleneck. The label was replaced by that observation at every prose site and in the equal-CPU and cluster table captions (the latter also corrected: its own data shows Prisma flat across 1/2/4 cores, so "Prisma requires roughly four cores to approach pg" was wrong—the four are cluster workers, not cores). Merging a now-redundant sentence offset the reframe, so the body is 11,443 and the total 14,990; the CPU-utilization measurements (" $\approx 110\%$ of one core") are unchanged.

A final round-6 clarification separated the *fact* that an earlier Prisma result did not reproduce on the re-run from the *hypothesis* that Prisma's version/architecture change caused it: the Discussion now states the non-reproduction as a fact and its cause as an explicitly hedged hypothesis (the vanished premium was Prisma-specific and only Prisma changed architecture, so its Rust-free rewrite is the likeliest driver, but the re-freeze advanced the whole toolchain at once, so it cannot be isolated), and the intro/conclusion headline sentences now say the result was retired "on newer versions" rather than attributing it to "Prisma's Rust-free successor." Trimming the paragraph's lead-in and a parenthetical offset most of the hedge, taking the body to 11,447 and the total to 14,994.

A final round-6 terminology change replaced the over-claiming label "idiomatic" (~53 occurrences) with "documentation-selected" throughout, with lighter synonyms ("documented default", "default") where it read clunkily. The methodology already disclaimed the connotation ("Documentation order is a reproducible selection heuristic, not evidence that the chosen API is performance-optimal or the most common production choice"); this makes the terminology consistent with that definition. The "default-configuration" estimand name is unchanged; the generated table captions (sameplan, altloading, insert-dispersion) and the replication-package **METHODOLOGY.md** were updated to match. Dropping the now-redundant "used idiomatically" disclaimer clause took the body to 11,437 and the total to 14,984.

A final round-6 change simplified the statistical emphasis: the results subsection "Measurement stability and significance" was renamed "...and effect sizes" and rewritten to lead with geometric-mean ratios, bootstrap CIs, and practical significance (the narrowest adjacent step, 1.05, sits at the $\pm 5\%$ equivalence margin — statistically distinguishable yet practically marginal; the up-to-7 $\times$ spread is decisive), with the post-hoc adjacent-pair permutation/Wilcoxon/Bonferroni tests compressed to a secondary, descriptive robustness note. The conclusion, methodology, and threats sentences were aligned to the same hierarchy (effect sizes and intervals primary; adjacent-rank tests secondary and descriptive). No statistic was added or removed; the significance table (Table 6) stays in the main text. Compressing the p-value enumeration offset the added practical-significance prose, so the body is 11,435 and the total 14,982.

A follow-on round-6 change reported the layer $\times$ engine interaction by its magnitude rather than its p-value: the RQ2 paragraph now leads with the per-layer PostgreSQL $\div$ MySQL throughput

ratios (reads a narrow $\approx 1.0\text{--}1.6\times$ band, the insert scattering $1.6\text{--}3.2\times$ and reordering layers across engines), with the blocked permutation test demoted to confirming the interaction is nonzero (its floor p says nothing about size); the outcomes-table row foregrounds the effect ratios (Table S28). The added magnitude detail took the body to 11,444 and the total to 14,991.

A final round-6 change softened the novelty claims to match the (explicitly non-systematic) scoping search: the one bald claim, "the combination missing from prior art" (introduction), became "a combination not identified in our documented search," and "a design choice absent from every access-layer benchmark surveyed above" (related work) became "not found in the access-layer benchmarks surveyed above." The four-axis gap statement and the related-work positioning were already scoped ("In the sources we searched..."; "a structured scoping search (not a systematic review)"). Net +3 words; the body is 11,447 and the total 14,994.

A final round-6 change stated explicit library inclusion/exclusion criteria in the methodology (each layer is maintained, in common use, occupies a distinct taxonomy tier, and — for the portable tiers — runs against both engines), with the reasons the named candidates were omitted: PostgreSQL-only clients (Slonik, pg-promise) are non-portable, and Kysely is a query builder already represented by Knex. The full criteria and a per-candidate table are in the replication package (`experiments/METHODOLOGY.md`, not word-counted). The addition was offset by removing a redundant tier-classification clause (Drizzle's placement, already given in the layer list) and tightening the vendor-independence sentence, so the net change is +1 word: the body is 11,448 and the total 14,995.

A final round-6 change added a short **artifact-reproducibility table** (Supplement Table S31: version/commit and DOI, software and hardware requirements, the run commands, the time and resources, and which results are regenerated automatically from the archived raw data), with a one-line pointer in the Data Availability section. Both are outside the counted body — the table is a supplement float (uncounted) and Data Availability is back-matter, not one of the seven counted sections — so the declared total is unchanged at 14,995. The supplement now holds thirty-one tables (S1–S31); the table is placed last, so S1–S30 keep their numbers. The current release is v1.6.1 (DOI 10.5281/zenodo.21428215).

Highlights (5 bullets, each ≤ 85 characters) are in `highlights.tex`.